

csc226-Rcode

```
# Load factors
alpha <- c(0.7,0.75,0.8,0.85,0.9)

# Chaining = 1 + alpha
chaining <- 1 + alpha

# Linear probing (successful)
lp_success <- 0.5 * (1 + 1/(1 - alpha))

# Linear probing (unsuccessful)
lp_unsuccess <- 0.5 * (1 + 1/((1 - alpha)^2))

# Create data frame
data <- data.frame(alpha, chaining, lp_success, lp_unsuccess)

data
```

	alpha	chaining	lp_success	lp_unsuccess
1	0.70	1.70	2.166667	6.055556
2	0.75	1.75	2.500000	8.500000
3	0.80	1.80	3.000000	13.000000
4	0.85	1.85	3.833333	22.722222
5	0.90	1.90	5.500000	50.500000

```
plot(alpha, chaining, type="o", pch=16, col="blue",
      ylim=c(0, max(lp_unsuccess)),
      xlab=expression(alpha),
      ylab="Expected Number of Probes",
      main="Expected Runtime vs Load Factor")

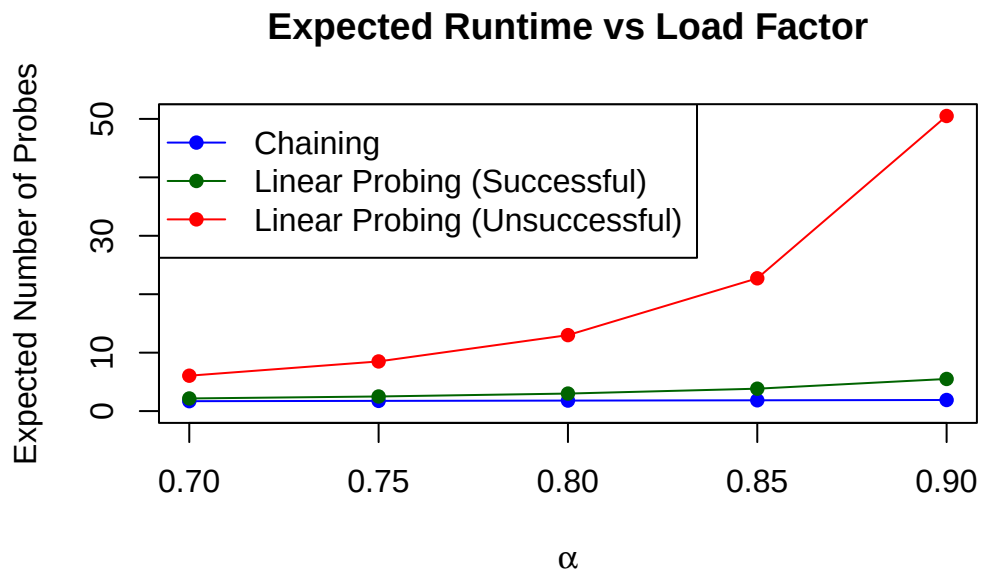
lines(alpha, lp_success, type="o", pch=16, col="darkgreen")
```

```

lines(alpha, lp_unsuccess, type="o", pch=16, col="red")

legend("topleft",
      legend=c("Chaining",
               "Linear Probing (Successful)",
               "Linear Probing (Unsuccessful)"),
      col=c("blue", "darkgreen", "red"),
      lty=1, pch=16)

```



```

# Read data
chaining <- read.csv("C:/Users/vuthu/OneDrive/Desktop/spring 2026/CSC226/assignment-notes/exe

linear <- read.csv("C:/Users/vuthu/OneDrive/Desktop/spring 2026/CSC226/assignment-notes/exe

plot(chaining$Load.Factor,
     chaining$Avg.Probes.Unsuccessful.Search,
     type="b",
     col="blue",
     pch=16,
     xlab="Load Factor (alpha)",
     ylab="Average Probes",
     main="Unsuccessful Search: Chaining vs Linear Probing")

```

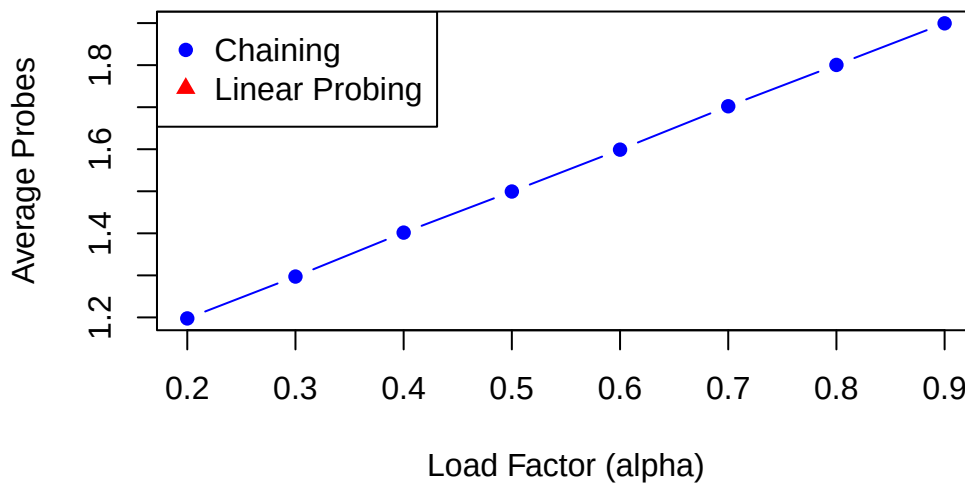
```

lines(linear$Load.Factor,
      linear$Avg.Probes.Unsuccessful.Search,
      type="b",
      col="red",
      pch=17)

legend("topleft",
      legend=c("Chaining", "Linear Probing"),
      col=c("blue", "red"),
      pch=c(16, 17))

```

Unsuccessful Search: Chaining vs Linear Probing



```

plot(chaining$Load.Factor,
      chaining$Avg.Probes.Successful.Search,
      type="b",
      col="blue",
      pch=16,
      xlab="Load Factor (alpha)",
      ylab="Average Probes",
      main="Successful Search: Chaining vs Linear Probing")

lines(linear$Load.Factor,
      linear$Avg.Probes.Successful.Search,

```

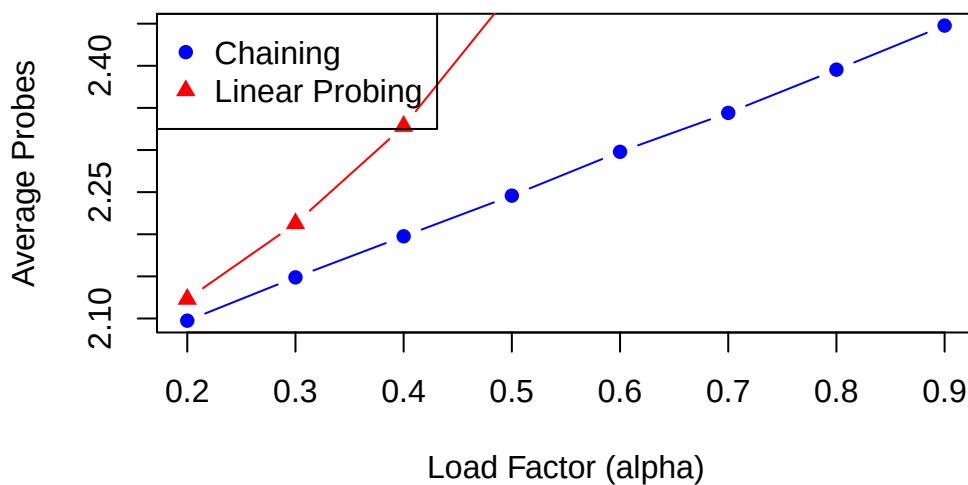
```

type="b",
col="red",
pch=17)

legend("topleft",
      legend=c("Chaining", "Linear Probing"),
      col=c("blue", "red"),
      pch=c(16, 17))

```

Successful Search: Chaining vs Linear Probing



```

plot(chaining$Load.Factor,
     chaining$Time.Unsuccessful.Search,
     type="b",
     col="blue",
     pch=16,
     xlab="Load Factor (alpha)",
     ylab="Time (seconds)",
     main="Runtime (Unsuccessful): Chaining vs Linear Probing")

lines(linear$Load.Factor,
      linear$Time.Unsuccessful.Search,
      type="b",
      col="red",

```

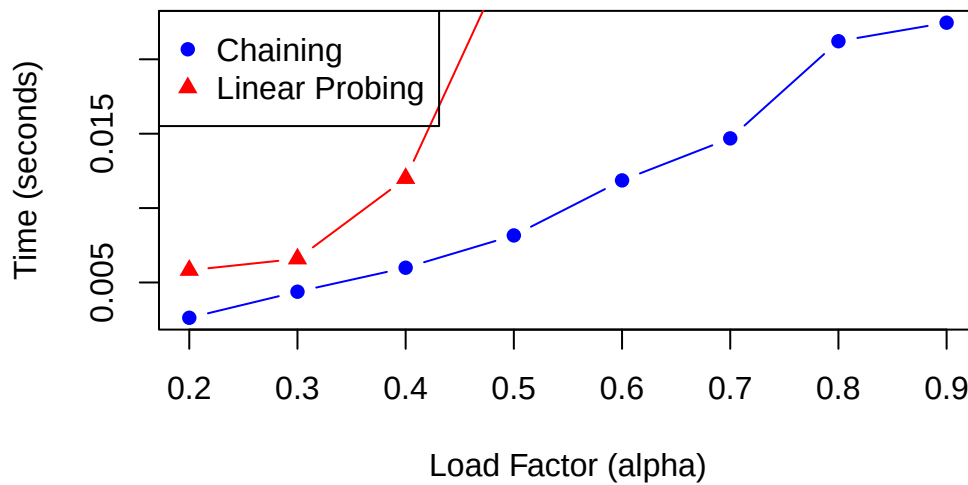
```

pch=17)

legend("topleft",
      legend=c("Chaining", "Linear Probing"),
      col=c("blue", "red"),
      pch=c(16, 17))

```

Runtime (Unsuccessful): Chaining vs Linear Probing



```

library(ggplot2)

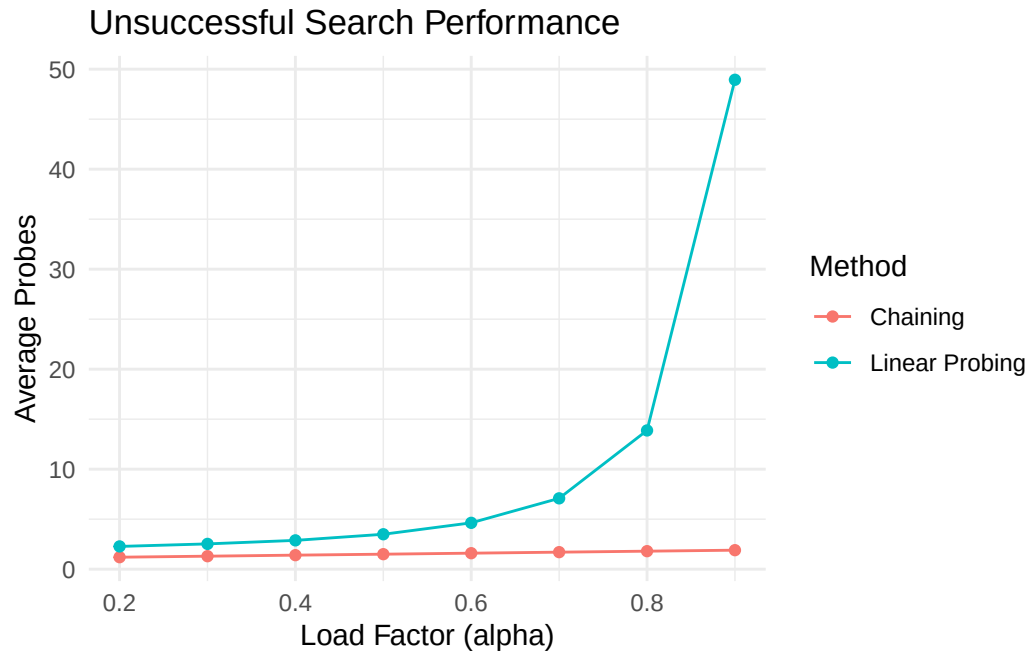
# Add method labels
chaining$Method <- "Chaining"
linear$Method <- "Linear Probing"

# Combine into one dataframe
data <- rbind(chaining, linear)

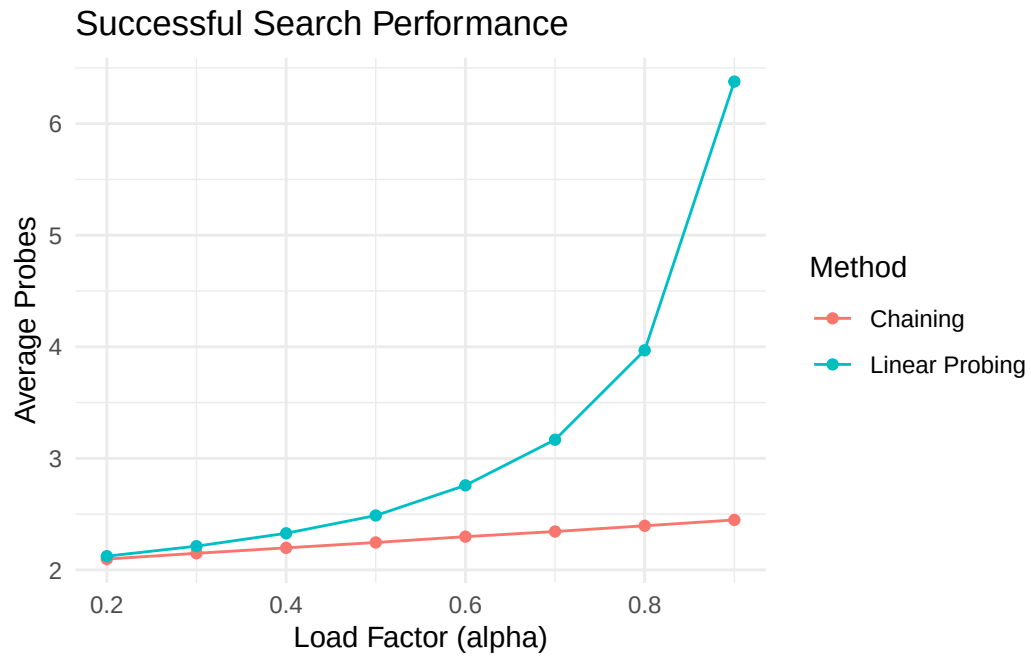
ggplot(data, aes(x=Load.Factor,
                  y=Avg.Probes.Unsuccessful.Search,
                  color=Method)) +
  geom_line() +
  geom_point() +
  labs(title="Unsuccessful Search Performance",

```

```
x="Load Factor (alpha)",
y="Average Probes") +
theme_minimal()
```



```
ggplot(data, aes(x=Load.Factor,
                  y=Avg.Probes.Successful.Search,
                  color=Method)) +
  geom_line() +
  geom_point() +
  labs(title="Successful Search Performance",
        x="Load Factor (alpha)",
        y="Average Probes") +
  theme_minimal()
```



```
ggplot(data, aes(x=Load.Factor,  
                 y=Time.Unsuccessful.Search,  
                 color=Method)) +  
  geom_line() +  
  geom_point() +  
  labs(title="Unsuccessful Search Running Time",  
        x="Load Factor (alpha)",  
        y="Time (seconds)") +  
  theme_minimal()
```

